

Ohnesorge Number

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For a liquid drop of diameter d , where the fluid has density ρ , dynamic viscosity μ , and surface tension σ , the Ohnesorge number is defined as

$$\text{Oh} = \frac{\mu}{\sqrt{\rho\sigma d}} . \quad (1)$$

This dimensionless number is typically described as gauging the relative strength of viscous forces and surface tension forces. Here we derive Oh in a way that better highlights its physical significance.

An inviscid liquid drop that is weakly perturbed from its spherical equilibrium shape will undergo oscillations that persist for all time. These so-called *capillary oscillations* are driven by surface tension, which acts in such a way as to oppose increases in the surface curvature. Dimensional arguments or a full analysis of the fluid equations show that the normal-mode oscillation frequencies are proportional to

$$\omega_0 = \sqrt{\frac{\sigma}{\rho a^3}} , \quad (2)$$

where $a = d/2$ is the radius of the drop. For axisymmetric perturbations, the lowest normal-mode frequency is $\sqrt{8}\omega_0$, corresponding to quadrupolar oscillations.

When viscosity is introduced, the capillary oscillations are damped. In order to approach a physical interpretation of Oh, we now consider these damped oscillations in the broader context of damped harmonic motion. For a damped oscillator, one often considers the “quality factor,”

$$Q = \frac{E}{\Delta E} , \quad (3)$$

where E is the kinetic energy of the oscillations, and ΔE is the energy lost in one oscillation cycle due to damping. Suppose the drop is undergoing normal-mode oscillations with frequency $\omega \propto \omega_0$ and amplitude $a\eta$, where η is a small dimensionless number. The bulk of the oscillatory energy is contained in a spherical shell of thickness $\sim a\eta$ and mass $\delta m \sim \rho a^2(a\eta)$ (we are not concerned here with numerical factors). Thus, the kinetic energy of the oscillations scales as

$$E \sim \delta m (a\eta)^2 \omega^2 = \rho a^2 (a\eta)^3 \omega^2 . \quad (4)$$

From the Navier-Stokes equation,

$$\rho \frac{d\mathbf{u}}{dt} = -\nabla p + \mu \nabla^2 \mathbf{u} , \quad (5)$$

we see that the local rate of viscous dissipation per unit volume scales as $\mu \mathbf{u} \cdot \nabla^2 \mathbf{u}$. Given the velocity scale $|\mathbf{u}| \sim a\eta\omega$ and the spatial scale a over which the velocity varies, we expect that the kinetic energy lost per period ($2\pi\omega^{-1}$) for the whole drop is

$$\Delta E \sim \omega^{-1} [a^2(a\eta)] \left[\mu \frac{(a\eta\omega)^2}{a^2} \right] = \mu (a\eta)^3 \omega . \quad (6)$$

We now find

$$Q \sim \frac{\rho a^2 \omega}{\mu} . \quad (7)$$

Since $\omega \propto \omega_0$ (see eq. [2]), the quality factor scales as

$$Q \sim \frac{\sqrt{\rho a \sigma}}{\mu} , \quad (8)$$

which is essentially just Oh^{-1} .

We have arrived at the conclusion that Oh is of order the inverse of the quality factor of the oscillating drop. In other words, Oh^{-1} is roughly the number of periods of normal-mode oscillation over which the energy is reduced by a significant factor. Alternatively, we can say that Oh is the characteristic period of capillary oscillations ($\tau_{\text{cap}} \sim \omega_0^{-1}$), divided by the time scale over which viscous dissipation acts to reduce the energy ($\tau_{\text{vis}} \sim \rho a^2 / \mu$).